Studio Apartment Composition, Fit & Budget Logic

AID-LOG-006 | Revision 1

Purpose. Defines the project-level backend logic used by AIDOOi to convert Studio Apartment inputs into an integrated furnishing composition, coordinate functional zones, validate fit and circulation, apply total-project budget logic, and generate recommendations. Customer journey inputs are defined in AID-SPC-006.

> **Core sequence: Total Area -> Studio Size -> Sleeping Setup -> Sleeping Footprint -> Available Functional Space -> Zone Composition -> Fit -> Total Project Budget -> Recommendation.**

1. Project-Level Principles

1. A Studio Apartment is treated as one integrated furnishing project composed of coordinated functional zones, not as multiple independent rooms.

2. Studio identity remains authoritative regardless of total area. A large studio must not automatically be processed as separate Bedroom, Living Room and Dining Room projects.

3. The customer's Sleeping Setup is a primary project anchor: Standard Bed, Sofa Bed, or Space-Saving / Convertible Solution.

4. One product may serve more than one functional zone. Functional duplication should be avoided when a single product can appropriately perform multiple roles.

5. Functional coverage and fit have priority over furniture quantity.

6. Visual and furniture-based zoning should be preferred before dedicated divider products are introduced.

7. Additional area should first improve zoning, circulation, product scale and functional quality before increasing product quantity.

8. Style and Colour are project matching inputs under AID-DATA-009 and AID-DATA-007. They do not directly determine the numeric project budget.

9. Final Project Price equals the sum of Current Selling Prices of all selected products and recalculates dynamically when products are removed or substituted.

2. Studio Size Classification

Total Studio Area	Classification
< 25 m²	Compact
25-35 m²	Small
35-45 m²	Medium
45-60 m²	Large
60+ m²	Extra Large

Total Studio Area is supplied through AID-SPC-006. Area classification guides the project composition but does not override Sleeping Setup or actual layout constraints.

3. Functional Zone & Fit Logic

> **Base fit priority: Sleeping Solution -> Essential Storage -> Main Circulation -> Living Zone -> Dining Zone -> Media -> Secondary Seating -> Lighting -> Decor.**

Dependency	Project Logic
Sleeping Setup	Determines the sleeping footprint and directly influences the remaining functional space.
Standard Bed	May coexist with a separate sofa when the layout supports both. Preferred Bed Size references AID-CAT-004.
Sofa Bed	May serve both Sleeping and Living functions. A second sofa is not added by default.
Convertible Solution	May combine multiple functions; recommendation logic should value functional coverage rather than product count.
Dining Solution	May use a compact or convertible table when space is constrained.
Storage	Essential storage is prioritized early; compact studios should use vertical space efficiently.
Zoning	Use rugs, furniture orientation and placement before adding dedicated screens or divider products.

Fit Constraint Rule. AIDOOi must not force a fixed furniture list into the studio. When all proposed functions cannot coexist with acceptable circulation, the system should adapt the product solution and surface genuine constraints rather than over-furnish the project.

4. Composition by Studio Size

4.1 Compact Studio (< 25 m²)

Zone	Product	Qty	Preferred Fit Logic
Sleeping	Standard Bed / Sofa Bed / Convertible	1	Exactly according to selected Sleeping Setup.
Bedside	Bedside Table	0-1	Only with Standard Bed and where fit permits.
Living	Compact Sofa / Seating	0-1	Conditional; not duplicated when Sofa Bed provides living seating.
Living	Coffee / Side Table	1	Compact or space-saving.
Media	TV Unit	0-1	Compact; only with suitable placement zone.
Storage	Wardrobe	1	Compact / vertical, approximately 80-120 cm.
Dining	Compact Dining Table	0-1	Prefer 2-person or convertible solution.
Dining	Dining Chairs	0-2	According to dining solution.
Lighting	Lamps	1-2	Low-footprint / multi-purpose preferred.
Soft Furnishing	Rug	0-1	Small; only if functional zones remain clear.
Decor	Wall Art / Mirror / Decor	2-3	Wall-based preferred.
Greenery	Planter	0-1	Only with suitable floor zone.

Compact Rule: One product may serve multiple functions. Functional coverage has priority over furniture quantity.

4.2 Small Studio (25-35 m²)

Zone	Product	Qty	Preferred Fit Logic
Sleeping	Standard Bed / Sofa Bed / Convertible	1	According to selected Sleeping Setup.
Bedside	Bedside Table	1-2	For Standard Bed; based on side clearance.
Living	Sofa	1	Compact 2-Seater; omitted as duplicate when Sofa Bed already serves living.
Living	Coffee Table	1	Approximately 60-90 cm or compact / nesting.
Living	Side Table	0-1	Conditional on fit.
Media	TV Unit	1	Approximately 100-140 cm.
Storage	Wardrobe	1	Approximately 100-160 cm; efficient vertical use.
Dining	Dining Table	1	Typically 2-person; compact / extendable may be preferred.
Dining	Dining Chairs	2	Matched to table.
Lighting	Lamps	2	Distributed across functional zones.
Soft Furnishing	Rug	1	Small / Medium; may define Living Zone.
Decor	Wall / Surface Decor	3-4	Placement-zone driven.
Greenery	Planter	0-1	Conditional.

4.3 Medium Studio (35-45 m²)

Zone	Product	Qty	Preferred Fit Logic
Sleeping	Standard Bed / Sofa Bed / Convertible	1	According to selected Sleeping Setup.
Bedside	Bedside Tables	1-2	For Standard Bed; based on clearance.

Zone	Product	Qty	Preferred Fit Logic
Living	Sofa	1	Typically 2-3 Seater; avoid duplication with Sofa Bed.
Living	Accent Chair	0-1	Only when Living Zone supports it.
Living	Coffee Table	1	Approximately 70-110 cm.
Living	Side Table	0-1	Conditional.
Media	TV Unit	1	Approximately 120-180 cm.
Storage	Wardrobe	1	Approximately 140-220 cm.
Storage	Chest of Drawers	0-1	Conditional.
Dining	Dining Table	1	Typically 2-4 persons.
Dining	Dining Chairs	2-4	Matched to table.
Lighting	Lamps	2-3	Distributed across zones.
Soft Furnishing	Rug	1	Medium; usually defines Living Zone.
Decor	Wall / Surface Decor	3-5	Placement-zone driven.
Greenery	Planter	0-1	Conditional.

Zoning Rule: Prefer visual and furniture-based zoning before adding dedicated divider products.

4.4 Large Studio (45-60 m²)

Zone	Product	Qty	Preferred Fit Logic
Sleeping	Standard Bed / Sofa Bed / Convertible	1	According to selected Sleeping Setup.
Bedside	Bedside Tables	1-2	For Standard Bed; based on clearance.
Living	Sofa	1	Typically 3-Seater; avoid duplication with Sofa Bed.
Living	Accent Chair	1	Conditional on fit.
Living	Coffee Table	1	Approximately 90-130 cm.
Living	Side Table	1	Conditional on fit.
Media	TV Unit	1	Approximately 160-220 cm.
Storage	Wardrobe	1	Approximately 180-280 cm.
Storage	Chest of Drawers	1	Conditional on fit.
Dining	Dining Table	1	Typically 4 persons.
Dining	Dining Chairs	4	Matched to table.
Lighting	Lamps	3-4	Distributed by functional zone.
Soft Furnishing	Rug	1	Large; primarily defines Living Zone.
Decor	Wall / Surface Decor	4-6	Placement-zone driven.
Greenery	Planter	1	Suitable floor / corner zone.

Integrated Project Rule: Large Studio does not equal Bedroom + Living Room + Dining Room compositions added together.

4.5 Extra Large Studio (60+ m²)

Zone	Product	Qty	Preferred Fit Logic
Sleeping	Standard Bed / Sofa Bed / Convertible	1	According to selected Sleeping Setup.
Bedside	Bedside Tables	2	For Standard Bed where clearance permits.
Living	Sofa	1	Typically 3-4 Seater; avoid duplication with Sofa Bed.

Zone	Product	Qty	Preferred Fit Logic
Living	Accent Chairs	1-2	For a stronger Living Zone.
Living	Coffee Table	1	Approximately 100-140 cm.
Living	Side Tables	1-2	Based on seating layout.
Media	TV Unit	1	Approximately 180-240 cm.
Storage	Wardrobe	1	Approximately 220-300+ cm.
Storage	Chest of Drawers	1	Conditional on fit.
Dining	Dining Table	1	Typically 4-6 persons.
Dining	Dining Chairs	4-6	Matched to table.
Lighting	Lamps	3-5	Distributed across zones.
Soft Furnishing	Rug	1-2	May define Living and, where useful, Sleeping Zone.
Decor	Wall / Surface Decor	5-7	Placement-zone driven.
Greenery	Planter	1-2	Suitable floor / corner zones.

Studio Identity Rule: Studio identity remains authoritative regardless of total area. Extra space should improve zoning, circulation and product scale before increasing product quantity.

5. Studio Project Budget Logic

Budget is applied at whole-project level and covers the coordinated Sleeping, Storage, Living, Dining, Media, Lighting and Decor composition. Sleeping Setup acts as an important budget modifier because it can change both product count and functional overlap.

> **Budget relationship: Studio Size + Sleeping Setup -> Integrated Composition -> Project Budget Tier -> Style & Colour Matching -> Recommendation.**

The customer-facing project budget tiers are Essential, Value, Standard, Premium and Luxury. Numeric ranges below are target ranges for the complete Studio Apartment furnishing project.

Studio Size	Essential	Value	Standard	Premium	Luxury
Compact (<25 m²)	3,000-6,000	6,000-9,000	9,000-14,000	14,000-22,000	22,000+
Small (25-35 m²)	4,500-8,000	8,000-12,000	12,000-18,000	18,000-28,000	28,000+
Medium (35-45 m²)	6,000-10,000	10,000-15,000	15,000-23,000	23,000-36,000	36,000+
Large (45-60 m²)	8,000-13,000	13,000-20,000	20,000-30,000	30,000-48,000	48,000+
Extra Large (60+ m²)	10,000-16,000	16,000-25,000	25,000-38,000	38,000-60,000	60,000+

Sleeping Setup Modifier. Standard Bed, Sofa Bed and Space-Saving / Convertible solutions may distribute project cost differently. A multi-functional product may appropriately receive a larger share of budget when it replaces multiple separate products.

> **Budget Allocation Priority: Sleeping Solution -> Essential Storage -> Living Function / Zone -> Dining Function / Zone -> Media -> Lighting -> Secondary Furniture -> Decor.**

Functional Budget Rule: Budget should follow functional coverage, not product count.

Target Range Rule: A Project Budget Tier defines a target spending range, not a requirement to spend the full amount. AIDOOi must not add unnecessary products or upgrade products solely to consume budget.

Product-level prices and tier logic reference AID-LOG-001. Final project price uses Current Selling Price.

6. Recommendation Rules

1. Treat the Studio as one coordinated project, not as independent room recommendations.
2. Resolve Sleeping Setup and its footprint before composing remaining functional zones.
3. Avoid functional duplication when one product can appropriately serve multiple functions.
4. Protect essential storage and main circulation before adding secondary furniture or decor.
5. Prefer furniture-based and visual zoning before dedicated divider products.
6. Use Style and Colour to rank products that already satisfy fit, availability and project-budget logic.
7. Allow Product Price Tier mixing inside a Project Budget Tier.
8. Recalculate the project total dynamically when products are removed or substituted.
9. Do not over-furnish larger studios merely because more floor area or budget is available.

7. Standard References

Reference	Standard
AID-SPC-006	Studio Apartment Path Standard
AID-DATA-007	Product Colour Standard
AID-DATA-009	Product Style Standard
AID-LOG-001	Product Price Range Standard
AID-CAT-001	Seating Standard
AID-CAT-002	Sofas Standard
AID-CAT-003	Tables Standard
AID-CAT-004	Beds Standard

Reference	Standard
AID-CAT-005	Storage & Casegoods Standard
AID-CAT-006	Lighting Standard
AID-CAT-007	Rugs & Soft Furnishings Standard
AID-CAT-008	Decor & Accessories Standard

Revision 1 status: Approved MVP project logic. Numeric thresholds and composition preferences may be revised later using observed inventory, customer behavior and recommendation performance.